

10 (a) Stomata have an important role in photosynthesis and transpiration.

Figure 10.1 shows the formula for calculating the stomatal index for an area of leaf.

$$\text{stomatal index} = \frac{\text{number of stomata}}{\text{number of stomata} + \text{number of epidermal cells}} \times 100$$

Figure 10.1

(i) Table 10.1 shows the stomatal index of the lower epidermis of the leaves of some crop plants.

Complete Table 10.1 for cotton and tomato using the formula shown in Figure 10.1.

Table 10.1

crop	number of stomata per unit area	number of epidermal cells per unit area	stomatal index
cotton	6		15.8
groundnut	9	36	20.0
mung bean	5	12	29.4
sorghum	8	20	28.6
tomato	7	25	

[2]

(ii) The Russian wormwood, *Artemisia gmelinii*, has a stomatal index of 13.8 for its lower leaf surface and 1.7 for its upper leaf surface.

Explain the advantages to the plant for this difference in stomatal index values.

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..... [2]

The air in the glasshouse was **not** completely sealed off from the external atmosphere so any difference in the carbon dioxide concentration between the glasshouse and external atmosphere will gradually decrease unless there are processes to maintain the difference.



[4]



(c) ATP and reduced NADP are products of the light-dependent stage of photosynthesis.

State where, in the chloroplasts, ATP and reduced NADP are produced **and** describe the roles of ATP and reduced NADP in the light-independent stage of photosynthesis.

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[Total: 11]